

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12412728
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Kusakabe; Ryuichi et al.

Electron beam inspection system

Abstract

There is provided an electron beam inspection system which can enhance the safety of the whole system if servo valves are deactivated in the event of a power failure or an emergency stop. The electron beam inspection system has a beam source, a stage mechanism, and a pump. The stage mechanism has a guide shaft, a slider, a first servo valve, a second servo valve, a first exhaust pipe, a second exhaust pipe, and an exhaust valve. The slider is movably supported to the guide shaft via a hydrostatic bearing and has a first pressure subchamber and a second pressure subchamber. The exhaust valve is mounted in the first exhaust pipe. When the servo valves are in operation, the exhaust valve is opened. When supply of electric power to the servo valves is ceased, the exhaust valve is closed.

Inventors:	Kusakabe; Ryuichi (Tokyo, JP), Kobayashi; Noriyuki (Tokyo, JP)
Applicant:	JEOL Ltd. (Tokyo, JP)
Family ID:	84489364
Assignee:	JEOL Ltd. (Tokyo, JP)
Appl. No.:	17/844120
Filed:	June 20, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220406560 A1	Dec. 22, 2022

Foreign Application Priority Data

JP	2021-102160	Jun. 21, 2021
----	-------------	---------------

Publication Classification

Int. Cl.: H01J37/20 (20060101)

U.S. Cl.:

CPC H01J37/20 (20130101); H01J2237/20278 (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2002/0030170	12/2001	McKell	251/129.04	F15B 9/09
2004/0145119	12/2003	Nakamura	277/409	F16C 29/025
2008/0302427	12/2007	Watanabe	137/488	F16K 31/0613
2011/0204255	12/2010	Ooae	250/453.11	G03F 7/70716

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
106383497	12/2016	CN	G05B 19/4147
112013003328	12/2019	DE	H01J 37/20
3220221	12/2016	EP	G05B 23/0256
H06204107	12/1993	JP	N/A
H1193911	12/1998	JP	N/A
H11183672	12/1998	JP	N/A
H11190306	12/1998	JP	N/A
2007101444	12/2006	JP	N/A
WO-2004032212	12/2003	WO	B82Y 10/00

OTHER PUBLICATIONS

English machine translation for JP-H11183672-A (Year: 1999). cited by examiner
English machine translation for JP-H06204107-A (Year: 1994). cited by examiner
English machine translation for JP-H1193911-A (Year: 1999). cited by examiner
English machine translation for WO-2004032212-A1 (Year: 2004). cited by examiner
English machine translation for JP-2007101444-A (Year: 2007). cited by examiner
English machine translation for CN-106383497-A (Year: 2017). cited by examiner
English machine translation for DE-112013003328-B4 (Year: 2020). cited by examiner

Primary Examiner: Kim; Robert H

Assistant Examiner: Kaliszewski; Alina

Attorney, Agent or Firm: The Webb Law Firm

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2021-102160 filed Jun. 21, 2021, the disclosure of which is hereby incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present invention relates to an electron beam inspection system.

2. Description of the Related Art

(3) An electron beam inspection system has a stage apparatus by which a sample is movably supported. To permit the sample to be accurately irradiated with an electron beam, the stage apparatus is required to move at improved accuracy. In recent years, in order to improve the accuracy of movement of such stage apparatus, it is considered to use a stage apparatus that employs a hydrostatic fluid bearing. In this fluid bearing, there is a gap between a guide shaft and a movable body. A fluid membrane is created in this gap by a lubricating fluid. The movable body is floated a distance equal to the thicknesswise dimension of the fluid membrane away from the guide shaft, thus accomplishing smooth movement of the movable body. JP A 11 190306 sets forth a technique regarding an air stage assembly which uses a gas as a lubricating fluid. Furthermore, in JP A 11 190306, there is described a technique of causing a thrust to act on a sliding table by supporting the sliding table to a horizontal securing member via a hydrostatic fluid bearing in a non-contact manner, controlling the air pressures in two air pressure chambers formed between the securing member and the sliding table by means of an air pressure control system, and utilizing the difference between the air pressures.

(4) However, in the technique set forth in JP A 11 190306, when servo valves for supplying air into the pressure chambers deactivate in the event of a power failure or an emergency stop, there arises the problem that the position of a slider for moving the stage cannot be controlled. Therefore, where the technique set forth in JP A 11 190306 is applied to an electron beam inspection system, if the servo valves deactivate, the slider may move into a position defeating the user's intention and the sample placed on the stage may come into contact with other device, thus damaging the sample or other device.

SUMMARY OF THE INVENTION

(5) In view of the foregoing problem, it is an object of the present invention to provide an electron beam inspection system which as a whole provides improved safety if the servo valves deactivate in the event of a power failure or an emergency stop.

(6) In order to solve the above problem and to achieve the object of the present invention, an electron beam inspection system is provided which comprises: an electron beam source for emitting an electron beam; a stage mechanism for providing movable support of a sample irradiated with the electron beam; and a pump for supplying air to the stage mechanism. The stage mechanism has a guide shaft, a slider, a first servo valve, a second servo valve, a first exhaust pipe, a second exhaust pipe, and an exhaust valve. The slider is movably supported to the guide shaft via a hydrostatic bearing and has a first and a second pressure subchambers. The first servo valve controls the pressure in the first pressure subchamber. Similarly, the second servo valve controls the pressure in the second pressure subchamber. Air discharged from the first servo valve passes through the first exhaust pipe. Air discharged from the second servo valve passes through the second exhaust pipe. The exhaust valve is mounted in the first exhaust pipe. When the first and second servo valves are in operation, the exhaust valve is opened. When power supply to these two servo valves is cut off, the exhaust valve is closed.

(7) An electron beam inspection system of the present invention has the advantage that the safety of the whole system can be enhanced if the servo valves deactivate in the event of a power failure or an emergency stop.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic block diagram of an electron beam inspection system associated with one embodiment of the present invention;
- (2) FIG. 2 is a plan view of a stage apparatus included in the electron beam inspection system of FIG. 1;
- (3) FIG. 3 is a cross-sectional view of one stage mechanism of the electron beam inspection system of FIG. 1;
- (4) FIGS. 4 and 5 illustrate an operational state occurring when the servo valves of the stage mechanism of the electron beam inspection system of FIG. 1 are inoperative;
- (5) FIG. 6 illustrates an operational state in which the slider of the stage mechanism of the electron beam inspection system of FIG. 1 has been retracted into its safety position;
- (6) FIG. 7 is an explanatory view of a lock mechanism for the stage mechanism of the electron beam inspection system of FIG. 1; and
- (7) FIG. 8 is a view similar to FIG. 7, but in which the lock mechanism is operative.

DESCRIPTION OF THE INVENTION

(8) Embodiments of the electron beam inspection system of the present invention are hereinafter described with reference to FIGS. 1-8, in which identical members are designated by identical reference numerals.

1. Embodiment

1-1. Configuration of Electron Beam Inspection System

(9) An electron beam inspection system associated with one embodiment of the present invention is first described by referring to FIG. 1, which schematically illustrates the system of this embodiment.

(10) The electron beam inspection system shown in FIG. 1 is generally indicated by reference numeral 1 and operates to irradiate a sample with an electron beam and to inspect a pattern on the sample. As shown, the inspection system 1 has an electron beam source 2 for generating the electron beam, a vacuum chamber 3, and a vibration isolated table 4. The vacuum chamber 3 has an interior 3a in which there are arranged a chuck 31 for holding the sample and a stage apparatus 40. The sample held by the chuck 31 is movably supported by the stage apparatus 40 whose configuration will be described in detail later. A sample exchange chamber 22 for exchanging the sample and a mask adjustment chamber 21 for adjusting the position of a mask are in communication with the interior 3a of the vacuum chamber 3.

(11) The electron beam source 2 further includes an electron gun for emitting an electron beam and an optics unit that includes an objective lens and condenser lenses. A high-voltage power supply (HVPS) 11 for supplying electric power to the electron gun and a lens power supply 12 for supplying electric power to the optics unit are connected to the electron beam source 2.

(12) Furthermore, the electron beam inspection system 1 includes a lithography (litho) pattern data preparation (prep) section 13, a data transfer section 14, a lithography (litho) control section 15, a stage control section 16, a laser metrology section 17, an air servo control section 18, and a pump 19 for providing vacuum control.

(13) The lithography pattern preparation section 13 creates lithography pattern data about a lithography pattern to be drawn on a sample and outputs the created pattern data to the lithography control section 15 via the data transfer section 14. The lithography control section 15 is connected to the beam source 2 and to the stage control section 16. The lithography control section 15 controls the beam source 2 and the stage control section 16 based on the lithography pattern data acquired from the data transfer section 14. Regarding the beam source 2, the lithography control section 15 controls the beam impact point, the timing of irradiation, the electron beam system, and

others.

(14) The stage control section **16** is connected to the laser metrology section **17** and to the air servo control section **18**. The laser metrology section **17** emits a laser beam at a mirror **32** mounted on the chuck **31** and receives laser light reflected from the mirror **32**. Then, the laser metrology section **17** measures the positions of the chuck **31** and sample supported on the stage apparatus **40**, and outputs positional information to the stage control section **16**.

(15) Based on the positional information from the laser metrology section **17** and on the control signal from the lithography control section **15**, the stage control section **16** outputs a stage apparatus control signal to the air servo control section **18**, thus controlling the actuation of the stage apparatus **40**. The air servo control section **18** is connected to the stage apparatus **40** and controls the actuation of servo valves **51**, **52** (described later in connection with FIG. 3) of the stage apparatus **40**.

(16) The pump **19** is connected to the vacuum chamber **3** and to the stage apparatus **40** and operates to suck in air from inside the vacuum chamber **3** and to supply air into the stage apparatus **40**. The pump **19** is supplied with electric power from a power supply that is different from the high-voltage power supply **11** connected with the beam source **2** and also from the lens power supply **12**.

(17) The stage apparatus **40** is next described by referring to FIG. 2, which is a plan view of the apparatus **40**. This apparatus **40** has a first stage mechanism **41**, a second stage mechanism **42A**, and a third stage mechanism **42B** each of which has a slider **43** and a guide shaft **44** by which the slider **43** is movably supported.

(18) The guide shaft **44** of the first stage mechanism **41** is disposed parallel to a first direction X that runs parallel to a horizontal direction. The second stage mechanism **42A** is located at one end of the first stage mechanism **41** as viewed in the first direction X, while the third stage mechanism **42B** is located at the other end of the first stage mechanism **41** as viewed in the first direction X. The opposite ends of the guide shaft **44** of the first stage mechanism **41** are supported on their respective ones of the sliders **43** of the second stage mechanism **42A** and the third stage mechanism **42B**.

(19) The guide shafts **44** of the second stage mechanism **42A** and the third stage mechanism **42B** lie parallel to the horizontal direction and also to a second direction Y that intersects the first direction X. Furthermore, the guide shafts **44** of the second and third stage mechanisms **42A**, **42B** are supported on the vibration isolated table **4** (see FIG. 1). The chuck **31** (see FIG. 1) and a sample are placed on the slider **43** of the first stage mechanism **41**.

(20) Details of the configurations of the stage mechanisms **41**, **42A**, and **42B** are next described by referring to FIG. 3. Since these stage mechanisms **41**, **42A**, and **42B** are identical in configuration, only the first stage mechanism **41** is described herein. In the following description, the first stage mechanism **41** will be simply referred to as the stage mechanism **41**. FIG. 3 is a cross-sectional view of the stage mechanism **41**. As shown, the stage mechanism **41** has the slider **43**, the guide shaft **44**, the first servo valve **51**, the second servo valve **52**, a flow adjusting valve **53**, a flow regulator **54**, and an exhaust valve **55**.

(21) The slider **43** is hollow and rectangularly shaped. A pressure chamber **46** and through-holes **50** are formed in the slider **43**. The through-holes **50** are formed at opposite ends of the slider **43** as viewed in the first direction X and in communication with the pressure chamber **46**. The through-holes **50** in the second stage mechanism **42A** and in the third stage mechanism **42B** are formed at opposite ends of the slider **43** as viewed in the second direction Y. The guide shaft **44** extends through the through-holes **50**.

(22) Each through-hole **50** has a surface opposite to the guide shaft **44**, and a groove **49** is formed in this surface. An air supply pipe **61** is connected to the groove **49**. Air is always supplied into the groove **49** from the pump **19** (see FIG. 1) via the air supply pipe **61**. An air film is formed between the inner surface of the through-hole **50** of the slider **43** and the outer surface of the guide shaft **44** because compressed air is supplied into the groove **49**. Consequently, the slider **43** is supported out

of contact with the guide shaft **44**. The groove **49** defines a hydrostatic bearing.

(23) A partition **45** is formed on the guide shaft **44**, and is a flange protruding substantially at right angles from the outer surface of the guide shaft **44**. The partition **45** is disposed in the pressure chamber **46** of the slider **43** and on the guide shaft **44**. Therefore, the pressure chamber **46** of the slider **43** is partitioned into a first pressure subchamber **47** and a second pressure subchamber **48** by the partition **45**. The pressure subchambers **47** and **48** are in communication with the grooves **49** via the through-holes **50**.

(24) In the first stage mechanism **41**, the first pressure subchamber **47** is formed around one end of the partition **45** as viewed in the first direction X, and the second pressure subchamber **48** is formed around the other end of the partition **45** as viewed in the first direction X. In the second stage mechanism **42A** and the third stage mechanism **42B**, the first pressure subchamber **47** is formed around one end of the partition **45** as viewed in the second direction Y. The second pressure subchamber **48** is formed around the other end of the partition **45** as viewed in the second direction Y. The first pressure subchamber **47** is connected to the first servo valve **51** via a first pipe **62**, while the second pressure subchamber **48** is connected to the second servo valve **52** via a second pipe **64**.

(25) The first servo valve **51** and the second servo valve **52** are connected to the pump **19** (see FIG. 1) via their respective supply tubes. The first servo valve **51** and the second servo valve **52** are connected to the air servo control section **18** and their actuation is controlled by the air servo control section **18**. The first servo valve **51** controls the air pressure in the first pressure subchamber **47** by supplying and discharging air into and from the first pressure subchamber **47** in response to a control signal from the air servo control section **18**. The second servo valve **52** controls the air pressure in the second pressure subchamber **48** by supplying and discharging air into and from the second pressure subchamber **48** in response to a control signal from the air servo control section **18**.

(26) The slider **43** is moved along the guide shaft **44** by creating a differential air pressure between the air pressure in the first pressure subchamber **47** and the air pressure in the second pressure subchamber **48** by the first servo valve **51** and the second servo valve **52**. In particular, by making the air pressure in the first pressure subchamber **47** higher than that in the second pressure subchamber **48** by the two servo valves **51**, **52**, the slider **43** is moved toward one end in the first direction X (i.e., to the left as viewed in FIG. 3). By making the air pressure in the second pressure subchamber **48** higher than that in the first pressure subchamber **47** by the two servo valves **51**, **52**, the slider **43** is moved toward the other end as viewed in the first direction X (i.e., to the right as viewed in FIG. 3). The two servo valves **51** and **52** maintain the slider **42** at rest by making equal the air pressures in the first pressure subchamber **47** and in the second pressure subchamber **48**.

(27) If the first servo valve **51** and the second servo valve **52** are deactivated in the event of a power failure or an emergency stop, supply lines for the valves are closed off. That is, if the operation of these valves **51**, **52** is deactivated, air supplied from the pump **19** is not supplied into the first pressure subchamber **47** and second pressure subchamber **48** via the first pipe **62** and second pipe **64**. When the servo valves **51** and **52** are out of operation, exhaust lines connected to a first exhaust pipe **63** and a second exhaust pipe **65** (both described later herein) are opened.

(28) The first exhaust pipe **63** and the second exhaust pipe **65** are connected to the first servo valve **51** and the second servo valve **52**, respectively. The second exhaust pipe **65** for the second servo valve **52** is open at all times.

(29) The first exhaust pipe **63** for the first servo valve **51** branches into two pipes, i.e., an exhaust pipe **63a** and an adjusting pipe **63b**. The flow adjusting valve **53** and the flow regulator **54** which together constitute an exhaust flow adjusting mechanism are mounted in the exhaust pipe **63a**. The exhaust valve **55** is mounted in the adjusting pipe **63b**.

(30) When electric power is supplied to the exhaust valve **55**, it is opened. Conversely, when the supply of electric power is cut off, the valve **55** is closed. Therefore, when the stage mechanism **41**

is in normal condition, i.e., the first servo valve **51** and the second servo valve **52** are in operation, the exhaust valve **55** is opened. Air exhausted during operation of the first servo valve **51** is expelled from the exhaust valve **55**. When the stage mechanism **41** is out of operation, i.e., power supply to the first servo valve **51** and the second servo valve **52** ceases, the exhaust valve **55** is closed.

(31) When the pressure in the exhaust pipe **63a** reaches a given level, the flow adjusting valve **53** and the regulator **54** are opened. Then, the adjusting valve **53** and regulator **54** adjust the flow rate and pressure of air flowing through the exhaust pipe **63a**.

2. Example of Operation of Stage Mechanism

(32) An example of operation of the stage mechanism **41** of the above-described configuration when it is deactivated because of a power failure or an emergency stop is next described by referring to FIGS. 4-6. FIGS. 4 and 5 illustrate an operational state in which the stage mechanism **41** is out of operation and its slider **43** has been retracted to its safety position.

(33) Referring to FIG. 4 as well as to FIG. 1, if the electron beam inspection system **1** suffers a power failure or makes an emergency stop, the air servo control section **18** is also deactivated, and the first servo valve **51** and the second servo valve **52** of the stage mechanism **41** are deactivated. As described previously, the supply lines for the first servo valve **51** and the second servo valve **52** are cut off. This shuts off the supply of air into the first pressure subchamber **47** and the second pressure subchamber **48** from the first pipe **62** and the second pipe **64** which are connected to the first servo valve **51** and the second servo valve **52**, respectively.

(34) The pump **19** is supplied with electric power from a power supply different from all of the high-voltage power supply **11** connected with the beam source **2**, the lens power supply **12**, and the power supplies for electrically energizing the first servo valve **51** and the second servo valve **52**. Therefore, if any other device is deactivated, the pump **19** permits supply of air into the vacuum chamber **3** and into the stage mechanism **41**. In consequence, the groove **49** constituting the hydrostatic bearing of the slider **43** is supplied with air from the pump **19** via the air supply pipe **61**.

(35) The grooves **49** are in communication with the first pressure subchamber **47** and the second pressure subchamber **48**, respectively, and so if the first servo valve **51** and the second servo valve **52** come out of operation, air **70** is supplied into the first pressure subchamber **47** and the second pressure subchamber **48** via the grooves **49**.

(36) If the supply of electric power to the first servo valve **51** and the second servo valve **52** ceases, their exhaust lines are opened. Therefore, the air in the first pressure subchamber **47** flows into the first exhaust pipe **63** via the first pipe **62** and the first servo valve **51**. The air in the second pressure subchamber **48** flows into the second exhaust pipe **65** via the second pipe **64** and the second servo valve **52**.

(37) If the supply of electric power to the exhaust valve **55** mounted in the first exhaust pipe **63** for the first servo valve **51** is stopped, the valve **55** is closed as shown in FIG. 5. Therefore, air expelled from the first exhaust pipe **63** for the first servo valve **51** flows into the adjusting pipe **63b** from the exhaust pipe **63a** in which the flow adjusting valve **53** and the flow regulator **54** are mounted, whereby the flow of air through the adjusting pipe **63b** is limited. The second exhaust pipe **65** for the second servo valve **52** is opened.

(38) This creates a difference between the flow rate of air expelled from the first exhaust pipe **63** for the first servo valve **51** and the flow rate of air expelled from the second exhaust pipe **65** for the second servo valve **52**. In the present example, the flow rate of air expelled from the first exhaust pipe **63** for the first servo valve **51** decreases relative to the flow rate of air expelled from the second exhaust pipe **65** for the second servo valve **52**. As a result, the air pressure in the first pressure subchamber **47** connected to the first servo valve **51** becomes higher than that in the second pressure subchamber **48** connected to the second servo valve **52**.

(39) Because the first pressure subchamber **47** goes higher in inner air pressure than the second

pressure subchamber **48**, the slider **43** is moved toward one end as viewed in the first direction X (i.e., to the left as viewed in FIG. 6). Consequently, if the supply of electric power to the first servo valve **51** and the second servo valve **52** comes to a stop because of a power failure or an emergency stop, the slider **43** can be moved in a desired direction. Hence, the slider **43** can be retracted into a safety position. As a result, it is possible to prevent the slider **43** from moving in a manner contrary to user's intention; otherwise the slider **43** or the sample on it would touch and damage other device. Thus, the safety of the whole electron beam inspection system **1** can be enhanced.

(40) The flow adjusting valve **53** and the flow regulator **54** for adjusting the flow rate of exhausted air are mounted in the first exhaust pipe **63** for the first servo valve **51**. Therefore, it is possible to prevent the flow rate of the air discharged from the second exhaust pipe **65** for the second servo valve **52** from becoming abruptly greater than that of the air discharged from the first exhaust pipe **63** for the first servo valve **51**. Consequently, the differential pressure between the first pressure subchamber **47** and the second pressure subchamber **48** can be controlled and thus the moving speed of the slider **43** can be adjusted. As a result, the moving speed of the slider **43** can be prevented from increasing rapidly; otherwise, the slider **43** would contact other devices.

(41) FIGS. 7 and 8 illustrate a lock mechanism mounted in the stage mechanism **41**. As shown in FIG. 7, the lock mechanism, **82**, in the stage mechanism **41** operates to limit movement of the slider **43** when it has been retracted into its safety position (hereinafter may also be referred to as the retracted position). The lock mechanism **82** has a lock receptor **80** and a locking portion **81**. The lock receptor **80** is secured to the slider **43** and provided with an engagement hole **80a**.

(42) The locking portion **81** is affixed to the vibration isolated table **4** (see FIG. 1) on which the stage mechanism **41** is placed. The locking portion **81** is mounted at the retracted position of the slider **43**. An engagement pin **83** is mounted on the locking portion **81**.

(43) When the slider **43** moves into the retracted position as shown in FIG. 8, the engagement pin **83** faces the engagement hole **80a** in the lock receptor **80**. Since the engagement pin **83** protrudes from the locking portion **81**, the engagement pin **83** comes into engagement with the engagement hole **80a**. Consequently, when the slider **43** is in the retracted position, it can be locked by the lock mechanism **82**.

(44) Note that the lock mechanism **82** for locking the slider **43** is not restricted to the foregoing example. For example, the lock mechanism may be composed of a hook portion and a hook receptor that engages the hook portion mounted on one of the slider **43** and the vibration isolated table **4**, the hook receptor being mounted on the other. Various other lock mechanisms may also be utilized.

(45) It is to be understood that the present invention is not restricted to the embodiments set forth above and shown in the figures and that the invention can be implemented in variously modified forms without departing from the gist of the present invention delineated by the claims.

(46) In the above embodiment, the slider **43** of the stage mechanism **41** moves horizontally. The stage mechanism is not restricted to this example. For example, a stage mechanism whose slider **43** moves vertically may also be used.

(47) In the above embodiment, the present invention is described as one example of an electron beam inspection system. The invention is not restricted thereto. For example, the invention may also be applied to various other systems and apparatus including an electron beam lithography system, an electron microscope, an instrument for measuring samples using an electron beam, and an electron beam processing machine for processing samples using an electron beam.

(48) In the present specification, the word "parallel" is used. It is to be understood that it is not intended to mean "strictly parallel" but rather means "substantially parallel" as long as the intended function can be exhibited.

Claims

1. An electron beam inspection system comprising: an electron beam source for emitting an electron beam; a stage mechanism for providing movable support of a sample irradiated with the electron beam; and a pump for supplying air to the stage mechanism; wherein said stage mechanism has a guide shaft having a partition formed on a surface thereof, a slider having a pressure chamber that is partitioned into a first pressure subchamber and a second pressure subchamber by the partition and movably supported to the guide shaft via a hydrostatic bearing, a first servo valve operative to control the pressure in the first pressure subchamber, a second servo valve operative to control the pressure in the second pressure subchamber, a first exhaust pipe through which air from the first servo valve passes, a second exhaust pipe through which air from the second servo valve passes, and an exhaust valve mounted in the first exhaust pipe; wherein the exhaust valve is opened when the first and second servo valves are in operation and the exhaust valve is closed when supply of electric power to the first and second servo valves is cut off such that the first exhaust pipe is closed and the second exhaust pipe is open creating a higher pressure in the first pressure subchamber connected to the first servo valve than the pressure in the second pressure subchamber connected to the second servo valve, thereby causing the slider to retract into a safety position where the slider is prevented from moving in a manner contrary to a user's intention and the slider is moved relative to the partition formed on the guide shaft and partitioning the first pressure subchamber and the second pressure subchamber; wherein the slider of the stage mechanism floats out of contact with the guide shaft even when a supply of electric power is cut off; wherein said first exhaust pipe branches into an exhaust pipe and an adjusting pipe; wherein said exhaust valve is mounted in the exhaust pipe; and wherein an exhaust flow adjusting mechanism for adjusting the flow rate of air passing through the exhaust pipe is mounted in the adjusting pipe.
 2. The electron beam inspection system as set forth in claim 1, wherein said hydrostatic bearing is a groove formed in a surface of said slider opposite to said guide shaft, the groove being in communication with said first and second pressure subchambers, and wherein air is supplied into the groove from said pump via an air supply pipe.
 3. The electron beam inspection system as set forth in claim 1, wherein said pump is driven from a power supply different from power supplies that respectively energize said first servo valve and said second servo valve.
 4. The electron beam inspection system as set forth in claim 1, wherein said stage mechanism has a lock mechanism for restricting movement of said slider when it has moved into the safety position, the lock mechanism comprising a first locking portion provided on the slider that is configured to engage with a second locking portion affixed to a table on which the stage mechanism is placed.
-